

Question 10 – Interference of Sound Waves

(a) Calculate the wavelength of the sound

✔ 3 marks

Given:

- Frequency $f = 1700\text{ Hz}$
- Speed of sound in air $v = 3.40 \times 10^2\text{ m/s}$

$$\lambda = \frac{v}{f} = \frac{340}{1700} = 0.20\text{ m}$$

(b) Determine whether constructive or destructive interference occurs at P

✔ 4 marks

Distance from loudspeaker 1 = 1.80 m

Distance from loudspeaker 2 = 1.50 m

Wavelength $\lambda = 0.20\text{ m}$

$$\text{Path difference} = 1.80 - 1.50 = 0.30\text{ m}$$

Check whether the path difference is a multiple of $\frac{\lambda}{2} = 0.10\text{ m}$

$$\frac{0.30}{0.20} = 1.5 \Rightarrow \text{Destructive interference (odd multiple of } \frac{\lambda}{2})$$

✔ Conclusion: *Destructive interference occurs at point P.*

(c) Loudspeaker 1 is disconnected

✔ 2 marks

✔ Answer:

The **amplitude increases**, because **destructive interference no longer takes place**
Only **one source** remains, so there is **no cancellation**

🧠 Acceptable alternatives:

- Sound becomes **louder**
- Interference pattern disappears
- The amplitude is **no longer reduced** by destructive interference

🧠 Revision Tips

- **Interference** occurs when waves overlap from two coherent sources
- **Path difference** determines whether you get **maxima (constructive)** or **minima (destructive)**
- Remember:
 - Constructive: $\Delta d = n\lambda$
 - Destructive: $\Delta d = (n + 0.5)\lambda$